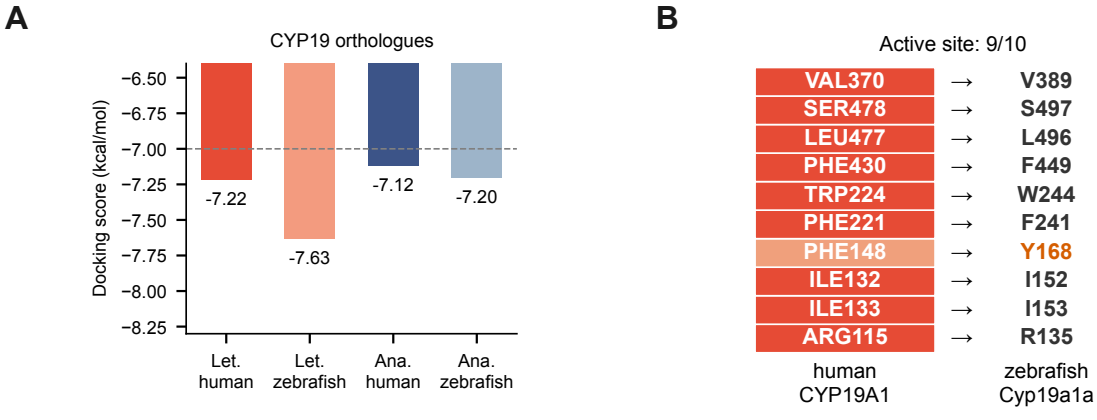
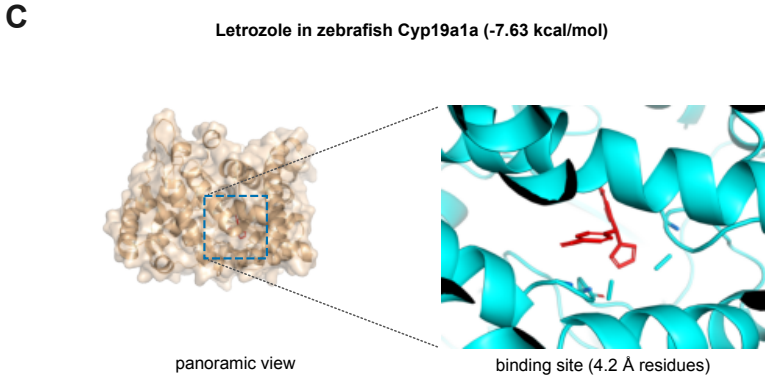
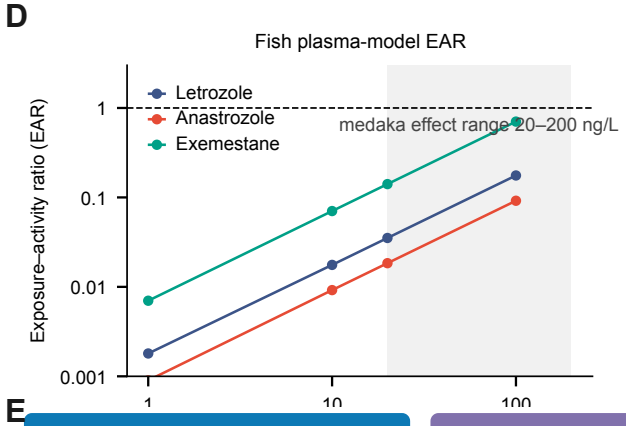
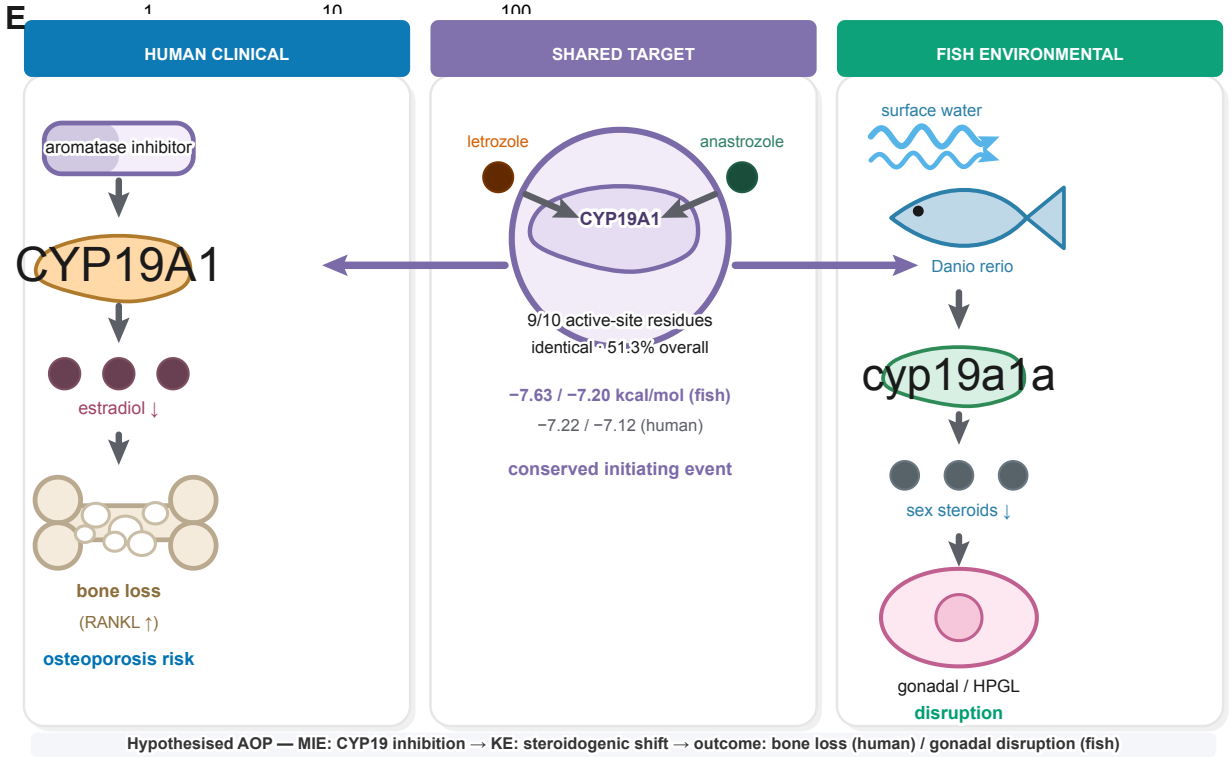

A
CYP19 orthologues
Docking score (kcal/mol)
−6.50
−6.75
−7.00
−7.25
−7.50
−7.75
−8.00
−8.25
-7.22
-7.63
-7.12
-7.20
Let. human
Let. zebrafish
Ana. human
Ana. zebrafish
B
Active site: 9/10
VAL370 → V389
SER478 → S497
LEU477 → L496
PHE430 → F449
TRP224 → W244
PHE221 → F241
PHE148 → Y168
ILE132 → I152
ILE133 → I153
ARG115 → R135
human CYP19A1
zebrafish Cyp19a1a


C
Letrozole in zebrafish Cyp19a1a (-7.63 kcal/mol)
panoramic view
binding site (4.2 Å residues)


D
Fish plasma-model EAR
Exposure–activity ratio (EAR)
Letrozole
Anastrozole
Exemestane
medaka effect range 20–200 ng/L
1
0.1
0.01
0.001
1
10
100


E
HUMAN CLINICAL
SHARED TARGET
FISH ENVIRONMENTAL
aromatase inhibitor
CYP19A1
estradiol ↓
bone loss
(RANKL ↑)
osteoporosis risk
letrozole
anastrozole
CYP19A1
9/10 active-site residues identical · 51.3% overall
−7.63 / −7.20 kcal/mol (fish)
−7.22 / −7.12 (human)
conserved initiating event
surface water
Danio rerio
cyp19a1a
sex steroids ↓
gonadal / HPGL
disruption
Hypothesised AOP — MIE: CYP19 inhibition → KE: steroidogenic shift → outcome: bone loss (human) / gonadal disruption (fish)